

Ben-Gurion University - Algorithms (202.1.3661)

Assignment 5: Divide and conquer.

Solve $T(n) = 2T(n/2) + O(n)$ via the Master Theorem.

Prove mergesort correct by induction on the array length n .

Show any comparison sort is $\Omega(n \log n)$ in the worst case.

Implement Dijkstra with a binary heap: $O((V + E) \log V)$.

Do not use AI tools such as ChatGPT or Claude for this assignment. All work must be your own.